

CSE/ECE 511 Computer Architecture: Quiz 1 (Set A)

Total Time: 30 Minutes

Maximum Marks: 20

Instructions:

1. Mobile phones, bags, wrist watches of any kind are NOT Permitted in the examination hall.
2. Write all the assumptions, if any, clearly. Only reasonable assumptions will be considered if any ambiguity is found in the question.
3. Institute's plagiarism policy will apply if any unfair means are observed.
4. Use of calculators is permitted.

Q1. [CO1] Two CPUs run the same program compiled for the same ISA. Both the CPUs are identical in nature (same frequency and same micro-architecture). However, the program which they are running is compiled by different compilers, say CMP-A and CMP-B. CPU-A runs the binary compiled by CMP-A, and CPU-B runs the binary compiled by CMP-B.

However, the number of instructions generated by both compilers is identical. Suppose that the ISA has two types of instructions: type-I and type-II. Type-I instructions take eight cycles to execute, and type-II instructions take 32 cycles. The binary running on the first CPU (CPU-A) has a 50-50 split between the two types of instructions. Given that CPU-A finishes execution in 20 seconds and CPU-B finishes execution in 10 seconds compute the split of instructions in the binary generated by CMP-B.

[10 marks]

Q2. [CO1] A CPU has a 3-level cache hierarchy with L1, L2 and L3 caches such that L1 sits closest to the CPU and L3 sits closest to the RAM. The hit rate, the cycles required to check if there is a hit, the cycles required to access the cache in case of a hit, and the RAM access latency are given below. Note that each level of the cache will have to pay the penalty to check for hits regardless of whether the access is a hit or not. Compute the expected data access latency for each level of this memory system. **[10 marks]**

Memory/Cache Type	Hit Rate	Hit Check Latency (Cycles)	Access Latency for hits (Cycles)
L1	70%	50	50
L2	80%	500	500
L3	90%	5,000	5,000
RAM	100%	0	60,000

CSE/ECE 511 Computer Architecture: Quiz 1 (Set B)

Total Time: 30 Minutes

Maximum Marks: 20

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3. Institute's plagiarism policy will apply if any unfair means are observed.
4. Use of calculators is permitted.

Q1. [CO1] Two CPUs run the same program compiled for the same ISA. Both the CPUs are identical in nature (same frequency and same micro-architecture). However, the program which they are running is compiled by different compilers, say CMP-A and CMP-B. CPU-A runs the binary compiled by CMP-A, and CPU-B runs the binary compiled by CMP-B.

However, the number of instructions generated by both compilers is identical. Suppose that the ISA has two types of instructions: type-I and type-II. Type-I instructions take eight cycles to execute, and type-II instructions take 32 cycles. The binary running on the first CPU (CPU-A) has a 50-50 split between the two types of instructions. Given that CPU-A finishes execution in 20 seconds and CPU-B finishes execution in 16 seconds, compute the split of instructions in the binary generated by CMP-B.

[10 marks]

Q2. [CO1] A CPU has a 3-level cache hierarchy with L1, L2 and L3 caches such that L1 sits closest to the CPU and L3 sits closest to the RAM. The hit rate, the cycles required to check if there is a hit, the cycles required to access the cache in case of a hit, and the RAM access latency are given below. Note that each level of the cache will have to pay the penalty to check for hits regardless of whether the access is a hit or not. Compute the expected data access latency for each level of this memory system. **[10 marks]**

Memory/Cache Type	Hit Rate	Hit Check Latency (Cycles)	Access Latency for hits (Cycles)
L1	75%	50	50
L2	85%	500	500
L3	95%	5,000	5,000
RAM	100%	0	60,000

CSE/ECE 511 Computer Architecture: Quiz 1 (Set C)

Total Time: 30 Minutes

Maximum Marks: 20

Instructions:

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3. Institute's plagiarism policy will apply if any unfair means are observed.
4. Use of calculators is permitted.

Q1. [CO1] Two CPUs run the same program compiled for the same ISA. Both the CPUs are identical in nature (same frequency and same micro-architecture). However, the program which they are running is compiled by different compilers, say CMP-A and CMP-B. CPU-A runs the binary compiled by CMP-A, and CPU-B runs the binary compiled by CMP-B.

However, the number of instructions generated by both compilers is identical. Suppose that the ISA has two types of instructions: type-I and type-II. Type-I instructions take eight cycles to execute, and type-II instructions take 32 cycles. The binary running on the first CPU (CPU-A) has a 50-50 split between the two types of instructions. Given that CPU-A finishes execution in 20 seconds and CPU-B finishes execution in 24 seconds compute the split of instructions in the binary generated by CMP-B.

[10 marks]

Q2. [CO1] A CPU has a 3-level cache hierarchy with L1, L2 and L3 caches such that L1 sits closest to the CPU and L3 sits closest to the RAM. The hit rate, the cycles required to check if there is a hit, the cycles required to access the cache in case of a hit, and the RAM access latency are given below. Note that each level of the cache will have to pay the penalty to check for hits regardless of whether the access is a hit or not. Compute the expected data access latency for each level of this memory system. **[10 marks]**

Memory/Cache Type	Hit Rate	Hit Check Latency (Cycles)	Access Latency for hits (Cycles)
L1	65%	50	50
L2	75%	500	500
L3	85 %	5,000	5,000
RAM	100%	0	60,000